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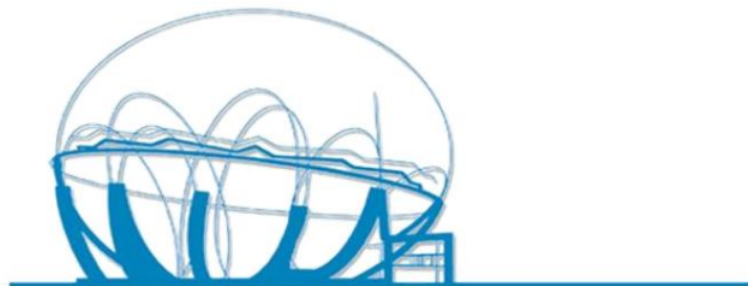
Deep Learning and Digital Communication Systems: Autoencoder with Feedback and Channel Estimation

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Chapter 1

General Introduction

The evolution of wireless communication systems towards higher data rates, lower latency, and increased reliability has pushed the boundaries of conventional signal processing techniques. As 5G and beyond communication systems face increasingly complex propagation environments, characterized by multipath fading, channel variability, and feedback imperfections, there is a growing need for intelligent and adaptive system designs.

In recent years, Deep Learning (DL) has emerged as a promising tool to revisit the physical layer design from a data-driven perspective. Notably, the use of autoencoders for end-to-end communication, as first introduced by O’Shea and Hoydis [1], has opened a new direction where modulation, coding, and detection can be jointly learned to match the channel conditions. However, many of the initial studies focus on idealized settings (e.g., AWGN, perfect CSI), leaving a gap between theory and practical deployment.

This project addresses the challenge of robust communication over realistic fading channels, with a focus on Rayleigh environments and imperfect CSI feedback. The central research question is:

“How can we design an autoencoder-based system that adapts to the imperfections of the wireless channel (e.g., noise, fading, feedback delay) and remains robust under practical constraints such as CSI uncertainty and power limitations?”

To address this question, we design, implement, and evaluate a deep autoencoder system that incorporates:

- A learnable feedback correction module for denoising CSI,
- Support for noisy, compressed, and delayed feedback links,
- A critical comparison with traditional systems under identical channel conditions,
- A diagnostic framework to analyze failures and guide future improvement (e.g., feedback-aware architectures, modulation constraints).

This report provides a detailed analysis of the system’s performance, its limitations, and future research directions to make autoencoder-based systems more deployable in next-generation wireless standards.

The full implementation of the proposed system, including training scripts, evaluation pipelines, and visualization tools, is publicly available on GitHub: <https://github.com/senhajiboutayna/DeepComm-Autoencoder-Realistic-Channels/tree/main>.

Chapter 2

Theoretical background

2.1 Rayleigh Channel

In wireless communication systems, multipath propagation means that the transmitted signal travels over multiple routes with varying delays and attenuations before reaching the receiver.

Multipath channels without a line-of-sight (LOS) component are statistically described by the **Rayleigh channel model**. This model predicts that the received signal's envelope has a Rayleigh-distributed amplitude [2], which is typical of heavily scattering environments in indoor or urban settings with a lot of nearby scatterers.

Mathematically, the channel gain h is modeled as a complex Gaussian random variable: $\mathbf{h} \sim \mathbf{CN}(0, 1)$ meaning that both the real and imaginary parts of h , denoted h_{Re} and h_{Im} , are independent and identically distributed (i.i.d.) Gaussian variables: $h_{Re}, h_{Im} \sim \mathcal{N}(0, 1/2)$.

The received signal y is typically represented as: $y = h * x + n$, where x is the transmitted signal, $n \sim \mathcal{CN}(0, N_0)$ is the additive white Gaussian noise (AWGN), and h varies randomly due to fading.

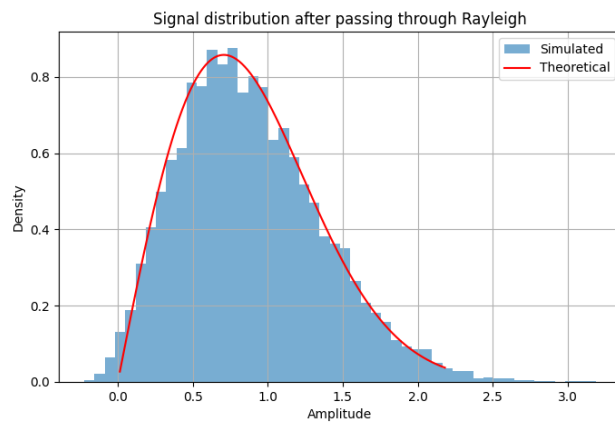


Figure 2.1: Distribution of the received signal amplitude under Rayleigh fading

Probability Density Functions :

The probability density function (PDF) of the channel amplitude $|h|$ is given by:

$$f_{|h|}(x) = \frac{x}{\sigma^2} \exp\left(-\frac{x}{2\sigma^2}\right), x \geq 0$$

where $\sigma = 1/\sqrt{2}$.

And the PDF of the channel power gain $|h|^2$ is :

$$f_{|h|}(x) = \exp(-x), x \geq 0$$

This exponential distribution plays a key role in the analysis of the average BER and capacity over Rayleigh channels.

Figure 2.1 illustrates the empirical distribution of the received signal amplitude after passing through a simulated Rayleigh fading channel, compared with the theoretical Rayleigh distribution.

The close match between the two validates our channel implementation and confirms that the simulated fading process follows the expected Rayleigh statistics.

2.2 Channel State Information

Channel State Information (CSI) refers to the knowledge of the characteristics of the propagation channel between the transmitter and receiver. CSI is typically represented by the complex gain h , which includes both amplitude fading and phase shifts caused by multipath propagation. Accurate CSI is essential for reliable data recovery, adaptive modulation, and beamforming.

The CSI h is critical for coherent detection, enabling the receiver to invert the channel effects via equalization :

$$\hat{x} = \frac{y}{\hat{h}} \approx x + \frac{n}{\hat{h}}$$

where $\hat{h}$ is the estimated CSI. Imperfect estimation ($\hat{h} \neq h$) introduces errors that propagate into the demodulated signal.

How is CSI Obtained in Practice ?

CSI is typically estimated by transmitting pilot symbols known to the receiver, which then performs channel estimation (e.g., via least squares or MMSE estimation) [3]. This estimated CSI is then either:

- Used locally at the receiver (CSIR), or
- Fed back to the transmitter (CSIT) to enable precoding and adaptive modulation.

However, this feedback link is often noisy, delayed, or limited in bandwidth, introducing imperfections that degrade the system's ability to adapt effectively.

Limitations of Practical CSI

In real systems:

- Estimation noise due to pilot interference or low SNR degrades CSI accuracy.
- Compression is required to reduce feedback bandwidth, introducing quantization errors.
- Delay between estimation and use introduces mismatch, especially in fast-fading environments.

This degradation is illustrated in Figure 2.2 :

- The blue distribution shows the signal envelope under perfect CSI, which closely follows a Rayleigh distribution.
- In contrast, the orange distribution under noisy CSI deviates significantly; noisy CSI introduces additional distortion, widening the signal distribution and increasing the probability of erroneous demodulation.

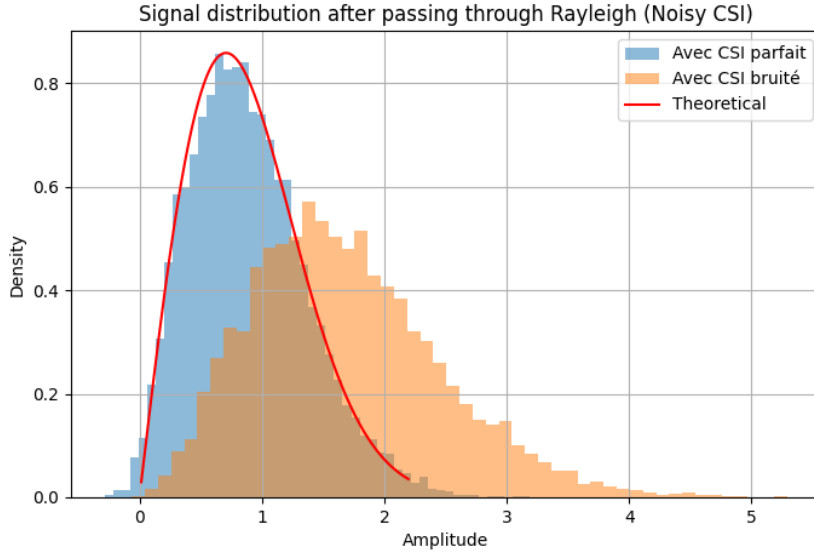


Figure 2.2: Signal amplitude distribution under perfect and noisy CSI conditions

These practical limitations of CSI, especially when used with deep learning-based communication systems, motivate the need for robust architectures that can operate under CSI uncertainty.

2.3 Overview of existing work : Deep Learning for Communication Systems

The application of deep learning to end-to-end communication systems has gained significant attention since the seminal work by O’Shea and Hoydis [1], who proposed modeling the entire transmitter–channel–receiver pipeline as a single autoencoder. In this framework, the neural network learns to jointly optimize modulation and coding schemes based on the actual channel conditions. This idea sparked a wave of research aimed at replacing or complementing traditional signal processing blocks with data-driven models [4].

Initially, autoencoders trained on Additive White Gaussian Noise (AWGN) channels demonstrated promising performance. However, generalizing these models to more realistic channels, such as Rayleigh or Rician fading, presented new challenges. To address this, many researchers extended the basic autoencoder to include channel models during training and, in some cases, explicitly incorporated Channel State Information (CSI) as an input to the encoder or decoder [4].

Recent research has highlighted the importance of training autoencoders over fading channels and introduced techniques to improve robustness, including CSI-aware decoding and the use of CSI estimation sub-networks [4]. In parallel, architectures based on recurrent and attention mechanisms have been proposed to exploit temporal correlations in time-varying channels [5]. Other approaches have integrated reinforcement learning to dynamically adapt the transmission policy in response to changing channel statistics [6].

A key challenge in this domain lies in the reliance on imperfect CSI in real-world scenarios, often due to noise, quantization, and feedback delays. While classical systems rely on pilot-based estimation and feedback mechanisms, deep learning offers alternative solutions such as CSI compression and neural-aided estimation to reduce overhead and improve robustness. Studies have, for instance, explored the use of variational autoencoders and reinforcement learning to optimize CSI feedback in massive MIMO systems, achieving significant reductions in bandwidth consumption without sacrificing detection accuracy [4], [5].

Beyond robustness, physical constraints like power normalization, spectral efficiency, or compatibility with standard modulation formats (e.g., QPSK, 16-QAM) have recently gained attention. Constraint-aware autoencoders that integrate power and constellation regularization have been proposed, demonstrating competitive performance under realistic channel and hardware limitations [4].

In summary, deep learning is reshaping the design of wireless communication systems by offering innovative solutions to challenges like fading, CSI acquisition, and real-time processing. The convergence of these works poses a central question: how to design a trainable, interpretable, and efficient communication system that operates reliably over dynamic and imperfect channels? Integrating deep learning with traditional signal processing methods, particularly through hybrid model-based and data-driven approaches, appears to be a promising direction for developing systems that are both high-performing and scalable. The remainder of this report will focus on addressing this question by exploring a novel autoencoder architecture that incorporates feedback, CSI correction, and paves the way for integrating physical constraints.

Chapter 3

Approach and Methodology

3.1 Global architecture of the system

Our communication system is a deep learning-based autoencoder, designed to efficiently transmit messages over a noisy channel with imperfect Channel State Information (CSI) feedback mechanism :

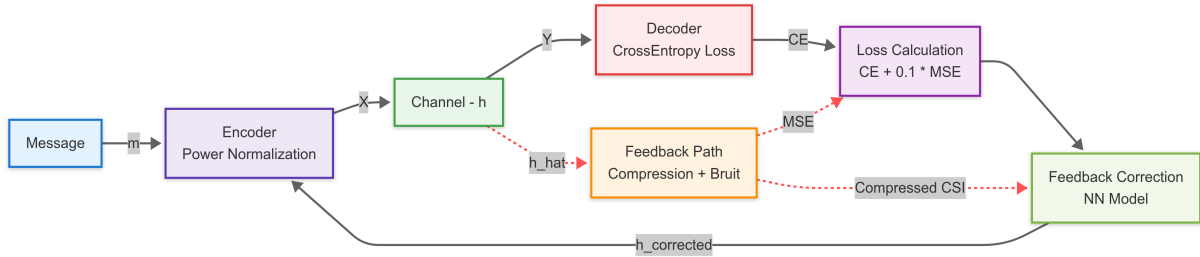


Figure 3.1: Global architecture of the system

Encoding and Power Normalization:

The process starts with a message $\mathbf{m}$, which is encoded into a continuous signal vector $\mathbf{X}$ by the encoder. During this step, power normalization is applied to ensure that the transmitted signal stays within energy constraints, which is essential for practical wireless systems.

Transmission over the Channel:

The encoded signal $\mathbf{X}$ is then sent through a communication channel with a complex gain $\mathbf{h}$. At the receiver, the signal undergoes distortions due to the channel and noise, resulting in a received signal $\mathbf{Y}$. Meanwhile, the channel estimate $\hat{h}$ is generated, but it is imperfect and modeled as $\hat{h} = h + \mathcal{N}(0, \sigma^2)$, where σ represents the noise level affecting the CSI feedback. This noisy CSI is one of the challenges the decoder needs to handle.

Decoding:

Finally, the received signal $\mathbf{Y}$ is passed through the decoder, which has been trained to reconstruct the original message by minimizing a CrossEntropy (CE) loss.

3.2 Adaptive CSI Feedback Mechanism

The red arrows in Figure 3.1 illustrate the flow of CSI feedback within our system.

3.2.1 Feedback path overview

The CSI is sent back to the transmitter in a compressed and noisy form, referred to as Compressed CSI. We account for three main practical challenges in modeling this feedback mechanism:

1. **Compression:**

In real-world systems, bandwidth limitations often require the CSI to be compressed before being sent back. To simulate this, we introduce a parameter called *compression_level*, which controls how much the CSI vector is reduced in size during the feedback process.

2. **Feedback Noise:**

The feedback link is not perfect and can introduce noise into the transmitted CSI. We model this imperfection using a feedback signal-to-noise ratio (denoted as *snr_feedback*). A lower SNR leads to a more distorted CSI at the transmitter, which can impair its ability to adapt effectively.

3. **Feedback Delay:**

In scenarios with high mobility, there is typically a time lag between the moment the CSI is measured and when it is actually used by the transmitter. This latency can lead to outdated information being used for adaptation. We simulate this effect by introducing a controlled feedback *delay* parameter in our system.

3.2.2 Neural Network-Based Correction

To address the inaccuracy of the CSI due to compression, noise, and delay, we integrate a dedicated neural network module, **FeedbackCorrection**.

Architecture Details :

The architecture of the FeedbackCorrection model is depicted in Figure 3.2. It follows an autoencoder design with an attention-based bottleneck.

The input to the network is the corrupted CSI vector of size *input_dim*, which is first processed by an encoder composed of two fully connected layers. Each linear layer is followed by Layer Normalization and a LeakyReLU non-linearity to stabilize training and ensure smooth gradients.

At the center of the network lies a bottleneck layer equipped with a Sigmoid-based attention mechanism. This mechanism allows the model to selectively amplify or suppress certain features in the latent space, helping it to better distinguish between informative components and noise.

The decoder then reconstructs the enhanced CSI vector through another two-layer stack, mirroring the encoder structure. The final layer outputs a vector of the same dimension as the input, passed through a Tanh activation to ensure the reconstructed values remain bounded.

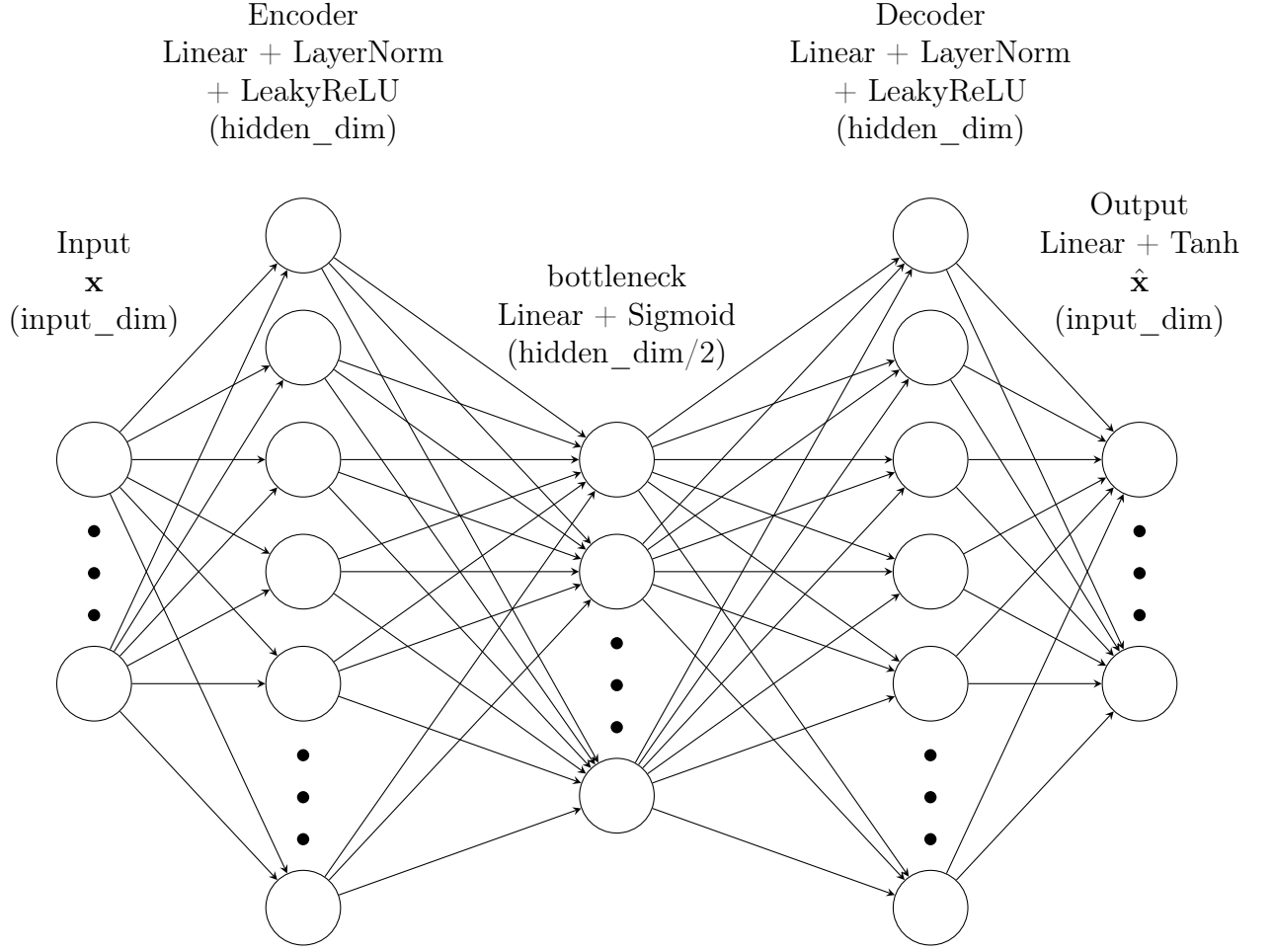


Figure 3.2: Architecture of the FeedbackCorrection neural network

Weight Initialization :

All linear layers in the model are initialized using Xavier uniform initialization, with zero biases. This choice promotes stable gradients and convergence during training, particularly for deep models operating in noisy environments.

3.2.3 Loss Function Design

Our system is trained using a composite loss function that balances two objectives: accurate message recovery and precise CSI reconstruction. Specifically, we combine the CrossEntropy loss (for decoding performance) with the Mean Squared Error (MSE) between the compressed CSI and its noisy estimate $\hat{h}$, as follows:

$$Loss = CrossEntropy(Y, m) + \lambda * MSE(Compressed\ CSI, \hat{h})$$

The hyperparameter λ is used to tune the trade-off between classification accuracy and feedback correction.

3.3 QPSK

To evaluate the performance of our autoencoder-based system, we compare it against a traditional communication scheme using **Quadrature Phase Shift Keying (QPSK)** modulation.

QPSK Transmission Chain:

The QPSK system used in our experiments follows the standard digital communication pipeline, with specific enhancements to handle noisy CSI feedback during equalization:

1. **Bit Generation:** A random sequence of binary bits is generated as the input data stream.
2. **QPSK Modulation:** Bits are grouped into pairs and mapped to complex symbols using the QPSK constellation:

$$s = \frac{1}{\sqrt{2}}[(2b_1 - 1) + j(2b_2 - 1)]$$

This ensures normalized average power for the transmitted symbols.

3. **Channel Transmission:** Modulated QPSK symbols are passed through one of three simulated channels: AWGN, Rayleigh and Rician Fading.

Channel gain h is applied to each symbol, and the receiver has access to a noisy CSI estimate $\hat{h}$.

4. **Equalization:** To mitigate the fading effect, the received signal is equalized using the noisy CSI feedback:

$$\hat{s} = \frac{y}{\hat{h}}$$

5. **Demodulation:** The equalized complex symbol is demapped to bits by detecting the quadrant in the complex plane:

$$\text{bit}_1 = \mathbb{I}(\Re(\hat{s}) > 0), \quad \text{bit}_2 = \mathbb{I}(\Im(\hat{s}) > 0)$$

6. **BER Computation:** The bit error rate (BER) is computed by comparing the demodulated bits to the original bitstream.

Chapter 4

Results and analysis

4.1 Experimental Setup Overview

The experimental evaluation of our system was conducted using a PyTorch-based simulation pipeline, modeling a full communication chain including encoding, channel effects, feedback, and decoding. All experiments were performed under these parameters:

- Message size (m) : 16
- Encoded vector length (n) : 7
- Signal-to-noise ratio (SNR) : 5 dB
- Training epochs : 50000
- Learning Rate : 0.0001
- Batch size : 128

These setups correspond to traditional communication settings, such as $(n, k) = (7, 4)$, where $k = \log_2(m)$.

Four configurations are compared:

- **Perfect CSI (blue):** where the channel state information is assumed to be exactly known at the receiver.
- **Noisy feedback CSI (orange):** where the CSI is corrupted by a noisy feedback channel and used directly without correction.
- **Noisy feedback with correction (green):** where the corrupted CSI is passed through a dedicated neural network to denoise it before being used for decoding.
- As a baseline, a **QPSK system (red)** is also included.

4.2 AWGN Channel with Feedback CSI

Figure 4.1 illustrates the Bit Error Rate (BER) performance of the proposed autoencoder system under an Additive White Gaussian Noise (AWGN) channel.

The **Perfect CSI** curve (blue line) represents the ideal scenario where the transmitter has

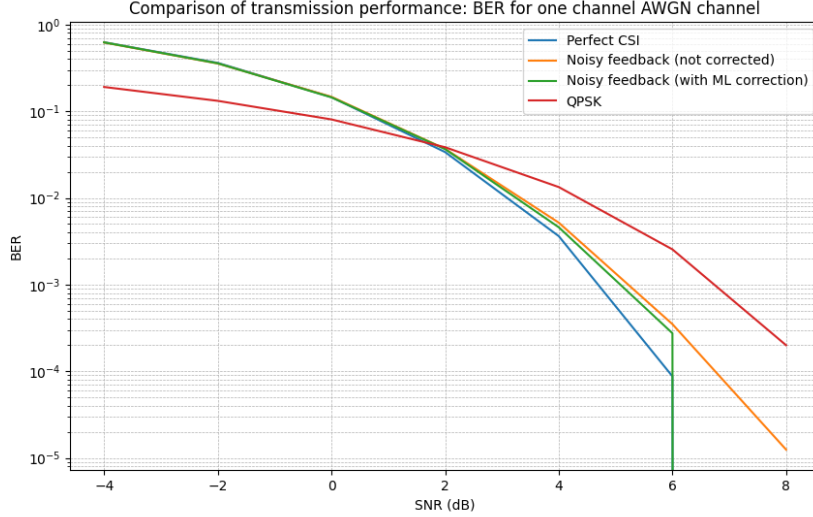


Figure 4.1: Comparison of transmission performance under AWGN channel

complete and accurate knowledge of the channel, serving as a lower bound for the achievable BER. As expected, this configuration demonstrates the best performance, achieving a BER of 10^{-5} at approximately 6 dB SNR.

In contrast, the **Noisy feedback (not corrected)** curve (orange line) shows a significant performance degradation. At an SNR of 6 dB, the BER for this scenario is considerably higher than with perfect CSI, indicating the detrimental effect of unmitigated feedback noise.

The **Noisy feedback (with correction)** curve (green line) highlights the effectiveness of integrating a machine learning-based correction mechanism for the noisy CSI feedback. This curve closely tracks the "Perfect CSI" curve, especially at lower SNRs. At an SNR of 6 dB, the system with correction achieves a BER comparable to the perfect CSI scenario, demonstrating the ML model's ability to effectively denoise and reconstruct the CSI.

Compared to the conventional **QPSK** (red) benchmark, the autoencoder-based system consistently achieves superior BER performance at higher SNRs, showcasing its inherent error correction capabilities and efficiency in an AWGN environment.

4.3 Rayleigh Channel with Feedback CSI

This section extends the performance analysis to a more realistic and challenging wireless propagation environment: the Rayleigh fading channel.

Unlike the static AWGN channel, Rayleigh fading introduces time-varying attenuation and phase shifts, making accurate channel estimation and robust communication significantly more complex.

4.3.1 BER Performance under Different CSI Conditions

Figure 4.2 compares the BER performance of our autoencoder-based communication system over a Rayleigh fading channel.

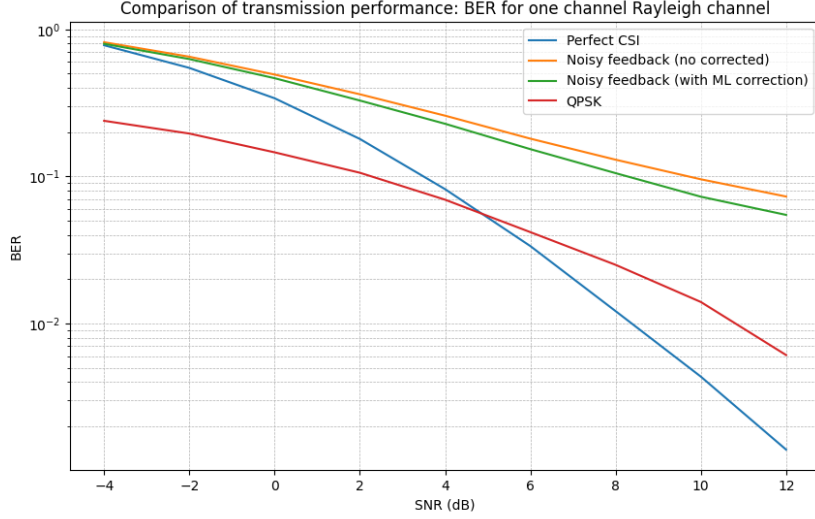


Figure 4.2: Comparison of transmission performance under Rayleigh channel

The Perfect CSI scenario performs best, achieving a steep BER decline as SNR increases. This validates the autoencoder’s ability to adapt its encoding and decoding strategies effectively when full channel knowledge is available.

However, the Noisy Feedback without correction case exhibits a significant performance degradation. The impact of noisy CSI feedback is even more pronounced in a Rayleigh fading channel compared to AWGN. This is because the CSI itself is inherently more volatile due to the fading process, and any additional noise in the feedback path severely corrupts the already dynamic channel information, leading to a much higher BER across all SNRs.

The ML-corrected CSI curve reveals a partial improvement over the raw feedback scenario at mid to high SNR values, confirming that the correction module is capturing some structure in the CSI noise. Yet, this improvement remains limited and inconsistent. Notably, the ML-corrected BER does not reach the performance of perfect CSI, and even performs worse than the QPSK baseline.

4.3.2 Identified problems

The results in Figure 4.2 reveal several critical challenges when applying the proposed ML-enhanced feedback system to Rayleigh fading channels. The issue is the substantial performance gap between the "Perfect CSI" and "Noisy feedback (with correction)" curves, which is more pronounced than observed in the AWGN channel.

Inadequate CSI Tracking in Dynamic Channels :

Rayleigh fading is time-varying [2], meaning the CSI changes rapidly.

The current ML correction model, even if effective at denoising, might not be adequately capturing or predicting these rapid fluctuations. This is a fundamental challenge for any feedback-based system in mobile environments.

Sensitivity to Training Data Realism :

The ML model's training data might not fully represent the diverse dynamics of realistic Rayleigh fading, including varying Doppler shifts and coherence times. If the training environment is too simplistic or doesn't cover the full operational range, generalization to actual fading conditions will be limited.

Limitations of Current ML Architecture :

The existing ML correction architecture might not be optimal for processing time-correlated data or for learning complex non-linear relationships in dynamic environments. Traditional feed-forward networks often struggle with temporal dependencies [7].

4.3.3 Possible Solutions and Future Work

Addressing these challenges requires a strategic shift in our approach. Future work will focus on :

- **Use of sequential models** such as LSTM or Transformer-based networks to model temporal dependencies in CSI feedback, especially in fast-fading scenarios.
- **Enhanced Training Data Strategies** by incorporating more diverse and realistic Rayleigh fading profiles into the training dataset, including variations in user mobility and channel coherence times, to improve the model's generalization capabilities.
- **Reinforcement learning** to adaptively adjust transmission strategies based on the quality of CSI, especially under constrained feedback links [6].

Chapter 5

General Conclusion and Perspectives

This project aimed to explore the feasibility and robustness of deep autoencoder-based communication systems over realistic wireless channels, with a particular focus on Rayleigh fading and imperfect CSI feedback. While the system performs acceptably under AWGN conditions, the results under Rayleigh fading have been disappointing, especially when CSI is noisy or delayed.

Despite these limitations, the project contributed to a deeper understanding of the complexity involved in making DL-based communication systems practical and interpretable.

A primary focus will be on improving the neural architecture by incorporating sequence modeling techniques such as RNNs or LSTMs, or leveraging attention mechanisms to better capture temporal dependencies in dynamic fading environments.

Furthermore, the encoder can be redesigned to become feedback-aware, adapting its behavior based on the quality or confidence of the CSI—potentially using reinforcement learning or probabilistic frameworks.

Another crucial step involves integrating physical constraints into the training process, such as enforcing standard modulation formats (e.g., QPSK, 16-QAM), average power limitations, and spectral efficiency. This could be achieved through regularization strategies, quantization-aware layers, or tailored loss functions.

Additionally, the training pipeline itself can be made more robust by incorporating realistic channel conditions through data augmentation, including Doppler effects and time-variant fading, and using curriculum learning to guide the model from simple to complex scenarios.

Finally, future evaluations will benefit from benchmarking against standardized communication systems under realistic setups, possibly including hardware-in-the-loop simulations or real channel traces.

The project does not end here; it opens the door for continued research into more adaptive, constrained, and interpretable systems. In the long term, such systems could play a key role in enabling intelligent and resilient communications in future wireless networks.

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